

WEEK 7-9: INTEGRATIVE PHYSIOLOGY

CARDIOVASCULAR SYSTEM

THREE MAJOR TYPES OF CARDIAC MUSCLE

- 1.) Atrial muscle
- 2.) Ventricular muscle
 - its thickness is used for power
- 3.) Specialized excitatory and conductive muscle fibers
 - Electricity used for muscle contraction

DEFINITION OF TERMS & IMPORTANT PRINCIPLES

1.) Cardiac Output (CO)

- Amount of blood ejected by heart per unit of time

FORMULA — same formula for VR

$CO = HR \times SV$

- At rest & normal value: 5L/m
- Max CO (non-athlete): 20L/m
- Max CO (athlete): 30L/m

- CO of the left heart = CO of the right heart

- CO right heart: pulmonary blood flow
 - Lower resistance, afterload, stroke work
- CO left heart: systemic blood flow
 - Same preload, heart rate, stroke volume, CO

2.) Venous return (VR)

- Amount of blood returning back to the heart per unit of time (overall)

3.) Central Venous Pressure (CVP)

- Equal to RAP; reflects amount of blood that returns to the heart (from IVC and SVC)

4.) Peripheral Venous Pressure (PVP)

- Pressure in the peripheral veins that drain the body tissues

5.) Stroke Volume (SV)

- Blood ejected by the ventricle per heartbeat

FORMULA

$SV = \text{End Diastolic Volume (EDV)} - \text{End Systolic Volume (ESV)}$

- Normal value: 70 mL

6.) Ejection Fraction (EF)

- Percentage of EDV that is actually ejected by the ventricle

FORMULA

$EF = SV / EDV$

- Normal value: 55%

7.) Stroke Work

- Work heart performs on each beat

8.) Systole

- Where the heart chambers are contracting and pumps the blood towards the periphery via arteries

9.) Diastole

- Where the heart chamber is in a state of relaxation, and fills with blood from the veins

PHASES OF CARDIAC MUSCLE AP

1.) Phase 0 (depolarization)

- Na channels are open
 - Na influx; MP is more positive

2.) Phase 1 (initial repolarization)

- Na channels closes; K channels slowly opens
 - K efflux

3.) Phase 2 (plateau)

- K channels closes; Cl channels open
 - Cl influx to compensate to K efflux

4.) Phase 3 (rapid repolarization)

- Cl channels closes; K channels open slowly

5.) Phase 4 (RMP)

- Averages about -90 mV

CARDIAC CYCLE

- Occur from the beginning of one heartbeat to the beginning of the next

- **Sinus node** - important for systole
- Starts with *diastole* (relaxation; filling of blood), then *systole* (contraction; ejection)

- When HR increases, the duration of each cardiac cycle decreases (contraction & relaxation phase)

REFRACTORY PERIOD

- Normal refractory period of ventricle: 0.25 to 0.30 second

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- About the duration of prolonged plateau AP
- Atria > Ventricle

EXCITATION-CONTRACTION COUPLING

- Mechanism by which AP causes the myofibrils of muscle to contract

- T tubules → longitudinal sarcoplasmic tubules release Ca ions → goes inside myofibrils → sliding of the actin and myosin filaments → muscle contraction

ELECTROCARDIOGRAM

TERM	DESCRIPTION
P Wave	Atrial depolarization
QRS Complex	Ventricular depolarization
T Wave	Ventricular repolarization
PR Interval	Start of atrial depolarization to start of ventricular depolarization; AP passes through AV node
QT Interval	Ventricular depolarization + ventricular repolarization
PR Segment	AV node conduction
ST Segment	Isoelectric portion; plateau phase (phase 2) of cardiac muscle AP

7 PHASES OF THE CARDIAC CYCLE

1.) Atrial contraction — “atrial systole”

- ECG: P wave
- Occurs during the distal third of diastole; nearing the end
- Not essential for ventricular filling
 - Ventricular pressure: increases slightly
 - Ventricular volume: increases slightly

2.) Isovolumic contraction

- ECG: QRS complex
- Heart sounds: S1 will be heard
- Ventricular pressure > atrial pressure
 - **AV valves will close**

- Semilunar valves are still closed; blood will not flow from LV to aorta

- Ventricular pressure: increases but ventricular pressure < aortic pressure
- Ventricular volume: remains the same

3.) Rapid ventricular ejection

- Semilunar valves open: blood will flow from LV to aorta

- Pressure difference is smallest in magnitude
- Atrial filling begins
 - Ventricular pressure: rapidly increases; ventricular pressure > aortic pressure
 - Ventricular volume: rapidly increases

4.) Slow ventricular ejection

- ECG: T wave will occur
- Aortic pressure decreases because of runoff of blood from large arteries to small arteries
 - Ventricular pressure: decreases
 - Ventricular volume: decreases

5.) Isovolumic relaxation

- ECG: T wave
 - Aortic valve and semilunar valve closes during this part of ECG
- Heart sound: S2 will be heard
- Ventricular pressure < aortic pressure
 - Ventricular pressure: rapidly decreases but ventricular pressure > atrial pressure
 - Ventricular volume: remains the same
 - **AV valves are still close**; no blood flow from atria to ventricles

6.) Rapid ventricular filling — during $\frac{1}{3}$ of diastole

- Heart sound: S3 may be heard
 - Ventricular pressure: rapidly increases; ventricular pressure < atrial pressure
 - Opening of the AV valves; blood rapidly flows from atrium to ventricles
 - Ventricular volume: rapidly increases

7.) Slow ventricular filling — during $\frac{2}{3}$ of diastole

- Longest phase of the cardiac cycle; dependent on HR
 - Ventricular volume: increase slowly

HEART SOUNDS

WEEK 7-9: INTEGRATIVE PHYSIOLOGY

AORTIC, PULMONARY, TRICUSPID, MITRAL — “Always Pray To Mary”

1.) S1

- Cardiac cycle: isovolumic contraction
- Longer, louder, lower in pitch
- Closure of AV valves

2.) S2

- Cardiac cycle: isovolumic relaxation
- Shorter
- Closure of semilunar valves

3.) S3

- Cardiac cycle: rapid ventricular filling
 - Ventricular gallop (always pathological)
- Congenital heart failure
 - Normal in children, young adults, pregnant patients
 - If in left side with CHF -> CVS morbidity, mortality

4.) S4

- Cardiac cycle: atrial contraction/systole
 - Atrial gallop (can be pathological)
- Myocardial infarction, hypertension
 - Stiff ventricles, vibration of ventricular wall during systole

PRELOAD & AFTERLOAD

1.) Preload

- Degree of tension on the muscle when it begins to contract

2.) Afterload

- The load against which the muscle exerts its contractile force

REGULATION OF HEART PUMPS

- Caused by Frank-Starling mechanism
 - Greater heart muscle is stretched during filling. greater the force of contraction, greater quantity of blood pumped into the aorta
 - Increased filling = increased afterload

EFFECT OF K IONS, CA IONS, & TEMPERATURE

1.) K ions

- Dilation of heart (flaccid)
- Slows HR
 - Cardiac weakness, abnormal rhythm, death

2.) Ca ions

- Spastic contraction
- Cardiac weakness

3.) Temperature

- Increased HR = fever

SINOATRIAL NODE — “sinus”

- Pacemaker of the heart
- Small, flattened, ellipsoid strip of specialized cardiac muscle
 - Located in the superior posterolateral wall of the right atrium, immediately and slightly lateral to the opening of the SVC
- Facilitates self-excitation through its fibers
 - Cause automatic rhythmical discharge and contraction
 - **Influx of Na and Ca**

AV NODE

- Located in the posterior wall of the right atrium, immediately behind the tricuspid valve
- Prevents the cardiac impulse to travel from atria to ventricles too rapidly
 - Allows time for the atria to empty their blood into the ventricles before ventricular contraction begins
- Divides into left and right bundle branches to Purkinje fibers
 - Rapid transmission compared to AV nodal fibers

PARASYMPATHETIC STIMULATION OF THE HEART

- CN responsible: CN10
- Neurotransmitters responsible: acetylcholine

SYMPATHETIC STIMULATION OF THE HEART

- Hormone responsible: norepinephrine

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- Increased rate of SA node discharge, conduction, level of excitability in all over the heart
- Increased force of contraction for both atrium and ventricle

ANS EFFECT ON THE HEART AND BLOOD VESSELS

	SYMPATHETIC		PARASYMPATHETIC	
	Effect	Receptor	Effect	Receptor
Heart Rate	↑	β_1	↓	M ₂
Conduction Velocity (AV Node)	↑	β_1	↓	M ₂
Contractility	↑	β_1	↓ (atria only)	M ₂
VASCULAR SMOOTH MUSCLE				
Skin, splanchnic	Constriction	α_1	Dilation (EDRF)	M ₃
Skeletal Muscle	Constriction	α_1	Dilation (EDRF)	M ₃
	Dilation	β_2	-	-
Veins	Constriction	α_1	-	-

WEEK 7-9: INTEGRATIVE PHYSIOLOGY

CIRCULATORY SYSTEM

PHYSICAL CHARACTERISTIC OF CIRCULATION

1.) Pulmonary circulation

- Supplies the lungs; **deoxygenated**

2.) Systemic circulation

- AKA greater circulation or peripheral circulation
- Supplies all organs away from the heart; **oxygenated**

FUNCTIONAL PARTS OF CIRCULATION

1.) Arteries

- Oxygen-rich blood except pulmonary arteries
- Pressure reservoir; transports blood under high pressure
 - Have strong vascular walls, and blood flows at a high velocity

2.) Arterioles

- Last small branches from arteries
- Act as control conduits through which blood is released into the capillaries
- Alters blood flow (constrict or dilate depending on the need)
 - Basically, it controls the BP

3.) Capillaries

- Exchange fluid, nutrients, electrolytes, hormones, and other substances between the blood and the interstitial fluid
 - Walls are thin, and pores are permeable to water and other small molecular substances

4.) Venules

- Collect blood from the capillaries and gradually travel into progressively larger veins
 - Drain blood from capillaries

5.) Veins

- Conduits for transport of blood from the venules back to the heart
- Major reservoir of extra blood — blood reservoir
 - Where ~60% of blood are stored
 - Can constrict or dilate
 - Help regulated CO by influencing VR
 - If hemorrhage happens, stored blood from veins will compensate to the blood loss

- Thin walls, low pressure than arteries
- Contains venous pump - propels the blood forward

VASCULAR DISTENSIBILITY

- Expressed as the fractional increase in volume per mmHg rise in pressure
 - % of volume change relative to original volume

FORMULA

Vascular Distensibility = Increase in Volume / Increase in Pressure X OG Volume

- Quantifies vascular compliance relative to pressure change

- All BVs are distensible (expandable)

- Allows to change volume with changes in pressure

1.) Arteries

- Less distensible; withstand pressure
 - Pulmonary veins: same distensibility to systemic veins
 - Pulmonary arteries: ~6X more distensible than systemic arteries
- Average out pressure pulsations, ensuring smooth, continuous blood flow in the small vessels
 - Prevents blood from flowing only during systole
 - Once the blood reaches capillaries, **flow is continuous with minimal pulsation (consistent)**

2.) Veins

- Most distensible
 - ~8X more distensible than arteries
- A slight increase in venous pressure will still allow vein to store 0.5 to 1 L of extra blood

VASCULAR COMPLIANCE — “capacitance”

- Total blood volume change per mmHg pressure rise

FORMULA

Compliance = Volume / Pressure

- BVs can be highly distensible but have low compliance if its volume is small
 - Systemic veins: 24X greater than arteries (8X)
 - Allows veins to act as blood reservoir
 - Arterial compliance ensures steady capillary flow

WEEK 7-9: INTEGRATIVE PHYSIOLOGY

VOLUME-PRESSURE CURVES

1.) Arterial system

- MAP $\approx$ 100 mmHg = $\sim$ 700 mL of blood
 - If volume drops to $\sim$ 400 mL, pressure falls to zero

2.) Venous system

- Normal value: 2000-3500 mL
- Several hundred of mL volume change $\rightarrow$ venous pressure shifts only to 3-5 mmHg
 - Explains why up to 0.5 L transfusion causes minimal circulatory disturbance

EFFECTS OF SYMPATHETIC STIMULATION & INHIBITION

1.) Sympathetic stimulation

- Increased vascular smooth muscle tone $\rightarrow$ higher pressure at the same volume (arteries & veins)
 - Allows the blood to shift from veins to heart to increase the cardiac output
 - Critical during hemorrhage: maintain circulation even with up to 25% blood loss

2.) Sympathetic inhibition

- Decreased tone $\rightarrow$ lower pressure at each volume

DELAYED COMPLIANCE — “stress-relaxation”

- Important for adaptation to extra blood during transfusion, and helps adjust circulation to blood loss during hemorrhage

1.) When volume suddenly increases

- Immediate elastic distension $\rightarrow$ rise in pressure
- Over minutes-hours: smooth muscle slowly relaxes, pressure falls towards normal
 - Basically, dilates when there's an increase in volume

2.) When volume decreases

- Vessels gradually contract, helping stabilize circulation

Feature	Arteries	Veins
Wall structure	Thick, strong, muscular	Thin, weaker
Distensibility	Low (resist expansion)	High ($\sim$ 8 $\times$ more than arteries)
Compliance ($\Delta V/\Delta P$)	Low	Very high ($\sim$ 24 $\times$ more than arteries)
Volume capacity	Smaller ($\sim$ 700 ml in systemic arteries)	Large ($\sim$ 2000–3500 ml in systemic veins)
Role in circulation	Maintain pressure & smooth out heart pulsations	Reservoir for blood storage
Response to pressure changes	Small volume change for given ΔP	Large volume change for small ΔP
Sympathetic stimulation	$\uparrow$ Tone $\rightarrow \uparrow$ arterial pressure	$\uparrow$ Tone $\rightarrow$ shifts blood to heart, supports circulation in hemorrhage
Pulmonary system	Pulmonary arteries: $\sim$ 6 $\times$ more distensible than systemic arteries	Pulmonary veins: similar to systemic veins
Adaptation (Delayed Compliance)	Limited effect	Strong stress-relaxation response $\rightarrow$ adjusts pressure after transfusion or hemorrhage

NORMAL PRESSURE VALUES

1.) Systolic pressure (SP): $\sim$ 120 mmHg

- influence 40% of HR

2.) Diastolic pressure (DP): $\sim$ 80 mmHg

- Influence 60% of HR

3.) Pulse pressure (PP)

FORMULA

Pulse pressure = SP - DP

- Normal value: $\sim$ 40 mmHg
 - If higher, hemorrhage is probably happening

DETERMINANTS OF PULSE PRESSURE

- Two major factors:

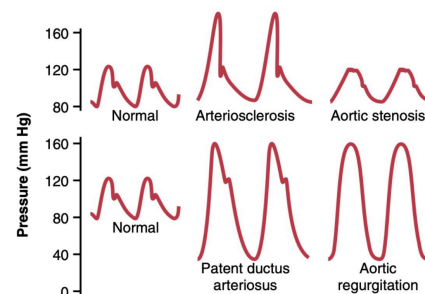
- SV output (increased SV $\rightarrow$ increased PP)
- Arterial compliance (decreased compliance $\rightarrow$ increased PP)

Pulse pressure = SV / Arterial Compliance

- Arteries stiffen with arteriosclerosis (less distensible)

- PP may double in old age

ABNORMAL PULSE PRESSURE CONTOURS — PAA



1.) Patent ductus arteriosus

WEEK 7-9: INTEGRATIVE PHYSIOLOGY

- Blood leaks from aorta to pulmonary artery -> low DP & widened PP

2.) Aortic stenosis

- Narrow valve (very constricted aorta) decreases SV -> decreases PP

3.) Aortic regurgitation

- Backflow into LV -> aortic pressure may fall to zero between beats, no dicrotic notch (incisura)

TRANSMISSION OF PRESSURE PULSES

- Pressure waves travel along arteries as a moving wave (continuous)

- Velocity of transmission:

- Aorta: 3-5 m/s (slow due to high compliance)
- Large branches: 7-10 m/s
- Small arteries: 15-35 m/s (fast due to low compliance)

- **Transmission velocity is 15X faster than blood flow velocity**

DAMPING OF PRESSURE PULSES

- Pulsation diminish in smaller arteries, arterioles, and capillaries

- Capillaries experience steady than pulsatile flow
- Ensures tissues receive continuous blood supply

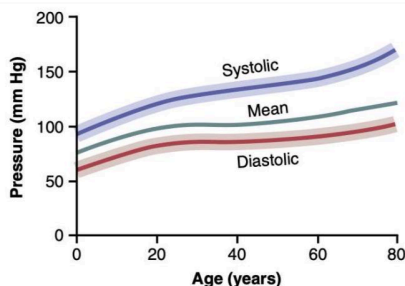
- Damping factors:

- Resistance - limits forward flow
- Compliance - requires more blood to raise pressure

FORMULA

Damping = Resistance X Compliance

MEAN ARTERIAL PRESSURE (MAP)



- Best indicator for tissue perfusion

- Average arterial pressure measured over time (mm by mm)

- Not the average of SP and DP because heart spends more time in DP at normal HR

- **Therefore, MAP is closer to DP**

FORMULA

$MAP \approx DP + \frac{1}{3} (SP - DP)$

- Normal value: closer to DP
- Very high HR: less time in diastole; MAP is the average of SP and DP

VENOUS PRESSURES

Central Venous Pressure (CVP) = Right Atrial Pressure (RAP)

- Normal: ~0 mmHg

- Abnormal increase of RAP: 20-30 mmHg in heart failure or massive transfusion
 - Difficulty in pumping
- Lower limit: 3 to ~5 mmHg (with strong pumping or severe hemorrhage)

- Determined by:

- Heart's pumping ability (strong heart with low RAP; weak heart with high RAP)
- Venous return (high RAP with more inflow)

- Factors that increase venous return (and RAP):

- Increased blood volume, venous tone (increased peripheral venous), and arteriolar dilation (rapid flow into the veins)

VENOUS RESISTANCE & PERIPHERAL VENOUS PRESSURE

- Large veins: very low resistance when distended

- Compression points (ribs, organs, neck, abdomen) have increased resistance

- PVP when lying down: 4-6 mmHg higher than RAP
 - High RAP (> 4-6 mmHg) - backup of blood, vein enlargement, rise in PVP

INTRA-ABDOMINAL PRESSURE

- Normal: 6 mmHg (recumbent/laying down)

- Can rise to 15-30 mmHg in pregnancy, obesity, ascites, tumors

WEEK 7-9: INTEGRATIVE PHYSIOLOGY

- Femoral venous pressure: must exceed intra-abdominal pressure to fight gravity for blood to flow toward the heart

GRAVITATIONAL/HYDROSTATIC PRESSURE

- BP increases 1 mmHg per 13.6 mm vertical distance due to gravity
- Standing person:
 - RAP remains ~0 mmHg (heart pumps excess)
 - Foot veins: ~90 mmHg due to hydrostatic column
 - Hand veins: ~35 mmHg (rib compression + hydrostatic effect)
 - Neck veins: collapse at 0 mmHg to prevent pressure rise/fall
 - Dural sinuses (inside the skull): can reach -10 mmHg (risk of air embolism if opened)

EFFECT ON ARTERIAL PRESSURE

- Gravity affects arterial and capillary pressures
 - E.g., MAP = 100 mmHg (at heart) + 90 mmHg (arterial pressure at feet) ≈ 190 mmHg
- Reported arterial pressure values are relative to heart level

VENOUS VALVES & VENOUS PUMP

- Muscle contractions compress veins -> blood is propelled forward (muscle/venous pump)
 - Walking/moving: keeps venous pressure in feet < 20 mmHg
 - Standing still: venous pressure in feet rises to 90 mmHg
 - **Hydrostatic hypotension** - Increased capillary pressure causes fluid leakage (leg swelling) and decrease in blood volume (10-20% lost in 15-30 mins), which could lead to possible fainting
 - Prevented by flexing leg muscles or bending knees to activate venous pump

VENOUS VALVE INCOMPETENCE & VARICOSE VEINS

- **Varicose veins** (bulging veins, esp. lower legs)

- From chronic high venous pressure (e.g., pregnancy, prolonged standing), which leads to overstretched veins, preventing valves to close

- Consequences:

- High venous & capillary pressure when standing
 - Chronic edema - impaired nutrient diffusion
 - Muscle pain/weakness, skin ulceration (insufficient blood supply), gangrene

- Management:

- Leg elevation, compression stockings/binders

BLOOD RESERVOIR FUNCTION OF VEINS

- Blood loss -> decreased arterial pressure -> baroreceptors (carotid sinuses, etc.) -> sympathetic activation -> vasoconstriction (prevents further blood loss) -> mobilizes stored blood
 - Circulation can still function even after ~20% blood loss because of veins

SPECIFIC BLOOD RESERVOIRS

1.) Spleen — can release ~100 mL of blood

- Venous sinuses: store whole blood
- Red pulp: stores concentrated RBCs trapped in trabeculae
 - Sympathetic activation -> contraction -> releases ~50 mL RBCs, raising hematocrit by 1-2% (amount of hemoglobin)
- White pulp: contains lymphoid tissue -> part of immune system (produces lymphocytes)

2.) Liver sinuses: several hundred mL

3.) Large abdominal veins: ~300 mL

4.) Subcutaneous venous plexus: several hundred mL

5.) Heart: ~50-100 mL (shrinks with sympathetic activity)

6.) Lungs: ~100-200 mL

FUNCTIONS OF MICROCIRCULATION

- Transport of nutrients to tissues & removal of cellular wastes

- Local control: small arterioles regulate blood flow; tissue conditions (nutrients, metabolites, H₂O) directly influences arteriolar diameter

WEEK 7-9: INTEGRATIVE PHYSIOLOGY

- Efficiency: ~10 billion capillaries with surface area of 500-700 m² ensures nearly every cell is within 20-30 µm of a capillary

ORGANIZATION OF MICROCIRCULATION

1.) Arterioles

- Highly muscular, diameter = 10-15 µm
- Branch into smaller arterioles (5-9 µm)

2.) Metarterioles (terminal arterioles)

- Intermittent smooth muscle fibers instead of continuous coat
- Can adjust depending on tissue metabolic needs

3.) Precapillary sphincters

- Rings of smooth muscle at capillary origins -> open/close to regulate blood flow into capillaries
- Can adjust depending on tissue metabolic needs

4.) Venules

- Larger than arterioles, weaker muscle coat, but can contract despite lower pressure

STRUCTURE OF THE CAPILLARY WALL

- Contains single **layer of endothelial cells + thin basement membrane**

- Thickness: ~0.5 µm
- Internal diameter: 4-9 µm (large enough for RBC)

CAPILLARY PORES & TRANSPORT PATHWAYS

1.) Intercellular clefts

- Narrow slits between endothelial cells (6-7 nm wide)
 - Account for ~1/1000 of capillary surface area
- Function: allow diffusion of water, ions, and small solutes

2.) Plasmalemmal vesicles (caveolae)

- Formed from caveolins + cholesterol + sphingolipids
- Function: endocytosis, transcytosis, and possible vesicular channel formation for macromolecule transport

SPECIALIZED CAPILLARY STRUCTURES BY ORGAN

1.) Brain (BBB)

- Tight junctions -> only very small molecules (H₂O, O₂, CO₂) can pass through

2.) Liver — sinusoid

- Large open clefts -> nearly all dissolved substances (plasma proteins) can pass

3.) GI tract

- Intermediate pore size between muscle and liver

4.) Kidney glomeruli

- Numerous fenestrae (small oval windows)
- Allow large amounts of small molecules/ions to filter (except plasma membrane)

FLOW OF BLOOD IN CAPILLARIES VASOMOTION

- Intermittent blood flow due to vasomotion (cyclical contraction/relaxation of metarterioles & precapillary sphincters)
 - **Main regulator: O₂ concentration in tissues**
 - Low O₂ -> sphincters stay open longer -> increased blood flow to deliver oxygen & nutrients
- Even though intermittent in capillaries, the average function of billions of capillaries result in a stable supply of nutrients and waste removal

EXCHANGE BETWEEN BLOOD & INTERSTITIAL FLUID

- Primary mechanism is diffusion through capillary membrane (from high to low)
 - Allow constant exchange
 - Proteins are the major plasma constituents that do not diffuse freely

PATHWAYS OF DIFFUSION

1.) Lipid-soluble substances (O₂, CO₂)

- Diffuse directly through endothelial cell membranes
- Very rapid transport due to full membrane permeability

2.) Water-soluble, non-lipid-soluble substances (Na, Cl, glucose, H₂O)

- Moves through intercellular clefts/pores

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- Despite pores being only 1/1000 of surface area, rapid thermal motion allows massive change
- E.g., water exchanges ~80 times faster than plasma flows linearly along the capillary

EFFECT OF MOLECULAR SIZE

Substance	Molecular Weight	Permeability
Water	18	1.00
NaCl	58.5	0.96
Urea	60	0.8
Glucose	180	0.6
Sucrose	342	0.4
Inulin	5000	0.2
Myoglobin	17,600	0.03
Hemoglobin	68,000	0.01
Albumin	69,000	0.001

- Capillary pore width: ~6-7 nm
 - Small molecules (H₂O, ions) - pass easily
 - Medium molecules (glucose, urea) - moderate passage
 - **Large molecules (albumin) - very limited passage**

CONCENTRATION GRADIENT PRINCIPLE

- Diffusion rate is directly proportional to concentration difference
 - O₂ (high in blood) - diffuses into tissues
 - CO₂ (high in tissues) - diffuses into blood
- Even small gradients can facilitate this principle

INTERSTITIUM & INTERSTITIAL FLUID

- **Interstitial** - space between cells ($\approx$ 1/3 of body volume)
- **Interstitial fluid** - fluid is derived from capillary filtration/diffusion, similar to plasma but with lower protein content
- **Structural Components**:
 - **Collagen fiber bundles**
 - Provides tensile strength
 - **Proteoglycan filaments (98% hyaluronic acid, 2% protein)**
 - Form a fine reticular network ("brush pile")
 - **Tissue gel**
 - Interstitial fluid entrapped in proteoglycan network

- Enables efficient diffusion-based exchange (not in bulk)
- Diffusion efficiency: 95-99% (as rapid as free fluid) -> allows fast exchange of H₂O, electrolytes, nutrients, O₂, CO₂, and wastes

• **Free fluid**

- Normally very little (< 1% of interstitial fluid)
- Flows in small rivulets along collagen or cell surfaces
- **Edema** - free fluid pockets expand greatly, leading to fluid accumulation independent of the proteoglycan matrix

FLUID FILTRATION ACROSS CAPILLARIES

- Determined by hydrostatic pressure and colloid osmotic pressure

- Balance is described by Starling forces

1.) **Capillary hydrostatic pressure (P_c)**

- Pushes fluid **out** of capillary
- $\approx$ 17-25 mmHg (varies by tissue)

2.) **Interstitial hydrostatic pressure (P_{if})**

- Positive - pulls fluid **into** capillary
- Negative - pulls fluid **out** of capillary
 - **Usually slightly negative** ($\sim$ -3 mmHg in subcutaneous tissue)
 - Intrapleural space: -8 mmHg
 - Joint synovial and epidural space: -4 to -6 mmHg
 - **Positive in encapsulated organs** (kidney, brain, eye, muscles)

3.) **Plasma colloid osmotic pressure (np)**

- Pulls fluid **into** capillary (mainly due to plasma proteins)

4.) **Interstitial colloid osmotic pressure (nif)**

- Pulls fluid **out** into interstitium

FLUID FILTRATION ACROSS CAPILLARIES

FORMULA

$$\text{NFP} = P_c - P_{if} - n_p + n_{if}$$

- **Positive NFP** - filtration into interstitium
- **Negative NFP** - absorption into capillaries

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- Capillary filtration coefficient (Kf)

- Reflects permeability & surface area of capillaries

FORMULA

Filtration = $K_f \times NFP$

CAPILLARY & INTERSTITIAL PRESSURES

- Role of lymphatic system

- **Acts as a scavenger system** - removes excess fluid, proteins, and debris

- Pumping of lymphatics helps maintain the slight negative interstitial pressure

COLLOID OSMOTIC PRESSURE (ONCOTIC PRESSURE)

- Plasma proteins (mainly albumin) create osmotic pressure

1.) Normal plasma colloid osmotic pressure: ~28 mmHg

- **19 mmHg** from proteins themselves
- **9 mmHg** from Donnan effect (extra osmotic effect of ions bound to proteins)

2.) Interstitial colloid osmotic pressure

- Small amounts of protein leak into interstitium via pores & vesicles

- Average interstitial protein concentration: ~ 3 g/dl (~40% of plasma)

- **Corresponding colloid osmotic pressure: ~8 mmHg**

STARLING FORCES GOVERN FLUID EXCHANGE

- Fluid exchange depends on the balance of hydrostatic pressures and colloid osmotic pressures across the capillary membrane

1.) Outward - arterial end (filtration)

- Outward forces:

- Capillary hydrostatic pressure = 30 mmHg
- Negative interstitial hydrostatic pressure = 3 mmHg
- Interstitial colloid osmotic pressure = 8 mmHg
- Total outward = 41 mmHg

- Inward forces:

- Plasma colloid osmotic pressure = 28 mmHg

- Total inward = 28 mmHg

- Net filtration pressure = 13 mmHg -> fluid moves OUT

- About 1/200 of plasma volume filters into interstitial fluid each pass

2.) Inward - venous end (reabsorption)

- Inward forces:

- Plasma colloid osmotic pressure = 28 mmHg
- Total inward = 28 mmHg

- Outward forces

- Capillary hydrostatic pressure = 10 mmHg
- Negative interstitial hydrostatic pressure = 3 mmHg
- Interstitial colloid osmotic pressure = 8 mmHg
- Total outward = 21 mmHg

- Net reabsorption pressure = 7 mmHg -> fluid moves IN

OVERALL FLUID BALANCE

- ~90% of filtered fluid at the arterial end is reabsorbed at the venous end

- ~10% enters the lymphatic system, which returns it to circulation

- This balance maintains normal fluid distribution between plasma and interstitial fluid

- **Essential to remember:**

- Filtration (arterial end) > reabsorption (venous end)
- Lymphatic system is critical to return excess fluid and prevent edema

STARLING EQUILIBRIUM CONCEPT

- Ernest Starling's principle: capillaries are normally in a state of near-equilibrium

- Ensures stable fluid balance between plasma and interstitial fluid

- Filtration $\approx$ reabsorption -> slight imbalance exists

- This slight excess filtration is drained by the lymphatic system

FORCES IN STARLING EQUILIBRIUM (MEAN VALUES)

1.) Outward force (28.3 mmHg total):

- Capillary hydrostatic pressure = 17.3 mmHg

- Negative interstitial hydrostatic pressure = 3 mmHg

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- Interstitial colloid osmotic pressure = 8 mmHg

2.) Inward force (28 mmHg in total)

- Plasma colloid osmotic pressure = 28 mmHg

3.) Net force = 0.3 mmHg -> slight net outward

- About 2 mL/min of fluid (excluding kidneys)
- This excess is normally returned via lymphatic system

CAPILLARY FILTRATION COEFFICIENT (Kf)

- Defines capacity of capillaries to filter fluid

- **Whole body Kf = 6.67 mL/min per mmHg imbalance**
 - Very small - brain, muscle
 - Moderate - subcutaneous tissue
 - Large - intestine
 - Extremely large - liver, kidney glomeruli

- Protein concentrations in interstitial fluid also vary:

- Muscle = 1.5 g/dL
- Subcutaneous tissue = 2 g/dL
- Intestine = 4 g/dL
- Liver = 6 g/dL

EFFECTS OF ABDOMINAL IMBALANCES

1.) Increase in capillary pressure -> increased filtration

-> if lymphatics cannot compensate -> edema

- E.g., 20 mmHg rise -> net filtration pressure increases to 20.3 mmHg (≈ 68X normal)

2.) Decrease in capillary pressure -> net reabsorption

-> blood volume increases at expense of interstitial fluid

REGULATION OF BLOOD PRESSURE

1.) Vasomotor area of the medulla

- Center responsible for regulation of HR and BP

- Lateral portion: excitatory area (increase BP & HR)
- Medial portion: inhibitory area (decrease BP & HR)

2.) Acute control

- ANS control, baroreceptors, chemoreceptors, low-pressure receptors, CNS ischemic response

3.) Long term control

- Renin-Angiotensin-Aldosterone system (RAAS)

- Slow: takes 20 minutes to take effect; takes several hours for optimal effects

- Activated when faster mechanism (e.g., baroreceptors) fail to regulate BP

- Also responsible for maintaining normal BP despite wide variation of salt intake

4.) ANS

- Sympathetic > parasympathetic

- To increase BP via ANS:

- Arteriolar vasoconstriction -> increase total peripheral resistance (TPR) -> increase BP
- Venous vasoconstriction -> increase VR -> increase CO -> increase BP
- Increase HR and SV via B1 receptors of the heart -> increase CO -> increase BP

4.) Baroreceptors

- Act fast; buffers minute-to-minute changes in BP

- Stretch receptors on the *carotid sinus* and *aortic arch*

- Increase BP -> increase stretch -> increase firing of CN 9 to nucleus tractus solitarius -> trigger parasympathetic response
 - Carotid baroreceptors: respond to increase/decrease in pressure from 50 mmHg to 180 mmHg
 - Aortic baroreceptors: respond to increase in pressure > 80 mmHg
 - Set point for MAP in vasomotor center: 100 mmHg

5.) Valsalva Maneuver

- Expiring against closed glottis -> increase intrathoracic pressure -> decrease VR -> decrease CO and BP -> sensed by baroreceptors -> increase HR

6.) Chemoreceptors

- Could also induce changes in BP

- Respond to low O₂, high CO₂ concentration whenever BP < 80 mmHg

- When stimulated, they activate vasomotor centers for systemic vasoconstriction -> increase TPR and BP

7.) Low-pressure receptors (cardiopulmonary baroreceptors)

- In the atria and pulmonary arteries

WEEK 7-9: INTEGRATIVE PHYSIOLOGY

- Detects "fullness" of vascular system (increase intravascular volume)

- In response to increased intravascular volume:

- **Increase atrial natriuretic peptide:** increase Na and H₂O excretion (where Na goes, H₂O follows)
 - In dehydration - increased ADH, angiotensin II, aldosterone, norepinephrine, and decreased ANP
 - ANP is released in response to increased atrial pressure from increased blood volume
- **ADH:** decrease urine output, more concentrated urine
- **Renal vasodilation:** increase urine output
- **Increase HR:** helps match VR with CO

8.) CNS ischemic response

- The vasomotor center itself responds directly to the ischemia during low BP

- Starts at BP < 60 mmHg and optimal at a BP = 15-20 mmHg

- The "last-ditch" stand:

- All systemic arterioles vasoconstrict severely EXCEPT for coronary vessels and cerebral vessels
- Directs blood to the brain

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM (RAAS)

1.) Baroreceptor reflex senses decrease in NaCl delivery in the macula densa of DCT

2.) It will stimulate JG cells and sympathetic nerve activity by β_1 receptor for renin secretion

3.) Converts liver Angiotensinogen to Angiotensin I

4.) Converts Angiotensin I to Angiotensin II in the lungs via Angiotensin-converting enzyme (ACE)

- Effects of Angiotensin II

- Maintains normal GFR (vasoconstricts efferent > afferent)
- Vasoconstricts systemic arterioles = increase TPR & BP
- Stimulates aldosterone secretion by adrenal cortex
 - Effects on kidney: Na and H₂O reabsorption, K secretion, H secretion

• Increase Na-H exchange in the PCT = contraction alkalosis

• Increase thirst and ADH secretion = increase intravascular volume

• Increase epinephrine, norepinephrine, cortisol secretion = vasoconstricts arterioles

VASOCONSTRICTORS	VASODILATORS
• Vasopressin: most potent vasoconstrictor • Serotonin: released because of blood vessel damage; causes arteriolar vasoconstriction; implicated in migraine • Endothelin: released by damaged endothelium • PGF and Thromboxane A₂ (TXA₂) • Norepinephrine, Epinephrine • Angiotensin II	• Prostacyclin (PGI₂): counteracts TXA₂ • Nitric Oxide (NO): vasodilates upstream blood vessels ◦ MOA: guanylate cyclase and cGMP ◦ Acetylcholine causes vasodilation by increasing production of NO in vascular smooth muscle ◦ Acetylcholine causes VASOCONSTRICTION instead of vasodilation whenever the endothelium is damaged due to decreased NO • PGE: vasodilators • Lactate, Adenosine found in muscles • Bradykinin and Histamine causes arteriolar dilation & venous constriction leading to increased filtration (local edema) • H⁺, CO₂, K, ANP

WEEK 7-9: INTEGRATIVE PHYSIOLOGY

MUSCLE BLOOD FLOW AND CO DURING EXERCISE

BLOOD FLOW IN SKELETAL MUSCLE

1.) At rest

- Blood flow = 3-4 mL/min/100 g muscle

2.) During extreme exercise (well-trained athletes)

- Blood flow can increase 25-50 fold = 100-200 mL/min/100 g muscle

BLOOD FLOW CHANGES DURING MUSCLE CONTRACTION

1.) Rhythmic contraction

- Blood flow fluctuates; decreases during contraction (vessel compression), increase after contraction

2.) Tetanic contraction

- Sustained compression and flow can be almost stopped (anaerobic), leading to fatigue

CAPILLARY RECRUITMENT IN EXERCISE

1.) At rest: many muscle capillaries are dormant (inactive)

2.) During exercise: nearly all capillaries are open, which:

- Decreases diffusion distance for O₂ and nutrients
- Increases capillary surface area 2-3 fold for exchange

TOTAL METABOLIC CONTROL OF BLOOD FLOW

1.) Primary mechanism: decreased oxygen levels -> arteriolar vasodilation

2.) Vasodilator factors released during activity

(increases blood flow as long as exercise continues):

- Lactic acid - metabolic waste of muscle after using ATP
- Adenosine (important but limited effect)
- ATP
- CO₂
- K ions

NERVOUS CONTROL OF MUSCLE BLOOD FLOW

1.) Sympathetic vasoconstrictor fibers

- Norepinephrine secretion decreases blood flow to $\frac{1}{2}$ - $\frac{1}{3}$ normal in resting muscle
- Helps maintain arterial pressure in shock/stress

2.) Adrenal medulla secretion during exercise

- Norepinephrine - vasoconstriction (via α -receptors)
- Epinephrine - slight vasodilation (via β -receptors)

THREE MAJOR CIRCULATORY ADJUSTMENTS IN EXERCISE

1.) Sympathetic nervous system activation (exercise)

- Strong stimulatory effects

2.) Increase arterial pressure

- Ensures adequate perfusion

3.) Increase in CO

- Supports high muscle demand

EFFECTS OF SYMPATHETIC ACTIVATION

1.) Heart effects

- Increased heart rate and contractility due to sympathetic drive
- Parasympathetic inhibition -> stronger cardiac output

2.) Arteriolar changes

- Vasoconstriction in most peripheral tissues (to help muscle during exercise)
- **Vasodilation in active muscles (local vasodilators dominate)**
- Blood is "redistributed" -> less to nonessential areas, more to muscle (up to 2 L/min extra flow)
 - Coronary and cerebral circulation spared -> protected from vasoconstriction

3.) Venous changes

- Venoconstriction increases mean systemic filling pressure
- Increases venous return to boost CO

SYMPATHETIC STIMULATION & ARTERIAL PRESSURE

- Mechanism for increased arterial pressure (what muscle blood flow depends during exercise)

WEEK 7-9: INTEGRATIVE PHYSIOLOGY

- Vasoconstriction in most tissue (except brain, heart, and active muscles)
- Stronger pumping activity of the heart
- Increased venous return due to venous contraction (venoconstriction)
- Stretches vessel walls, enhancing flow with local vasodilators

- Range of MAP increase:

- 20-80 mmHg depending on exercise conditions
 - 120/80 at rest

- Specific scenarios:

- Tense exercise with few muscles active -> vasoconstriction dominates -> MAP can rise to ~170 mmHg
- Whole body exercise (running, swimming) -> widespread vasodilation -> pressure rise only 20-40 mmHg
- Marathon runner: blood flow can increase 20 times than normal due to combined effects

Parameter	At Rest	During Exercise	Key Concept
Heart rate (HR)	Normal (60-80 bpm)	↑ significantly	Sympathetic activation + ↓ parasympathetic tone
Cardiac output (CO)	~5 L/min	↑ 4-7x (up to 25-30 L/min in athletes)	Increased HR + stroke volume
Stroke volume (SV)	Normal	↑ initially (then plateaus at high HR)	Stronger contraction + ↑ venous return
Arterial pressure (MAP)	~90-100 mmHg	↑ 20-80 mmHg	Sympathetic vasoconstriction + CO rise
Arterioles	Balanced tone	Vasoconstriction in nonactive tissues; vasodilation in active muscles	Redistribution of blood
Blood flow to muscles	15-20% of CO	80-85% of CO	Local vasodilators ($\downarrow O_2$, $\uparrow CO_2$, $\uparrow K^+$, lactic acid, adenosine)
Blood flow to brain	Constant (~750 ml/min)	Constant	Cerebral perfusion preserved
Blood flow to heart (coronary)	~250 ml/min	↑ 3-4x	Matches ↑ myocardial O_2 demand
Venous return	Baseline	↑ greatly	Venoconstriction + muscle pump
Capillary recruitment in muscle	Limited	Almost all open	Shorter O_2 diffusion distance, ↑ surface area

IMPORTANCE OF INCREASED CO DURING EXERCISE

- CO must rise proportionally to exercise intensity
 - **Normal: 55 mL per heartbeat**
- Ability of circulation to deliver oxygen and nutrients is as important as muscle strength
- High CO capacity = better endurance and athletic performance
 - Marathon runner with highest CO often set record-breaking time

GRAPHIC ANALYSIS OF CO CHANGES

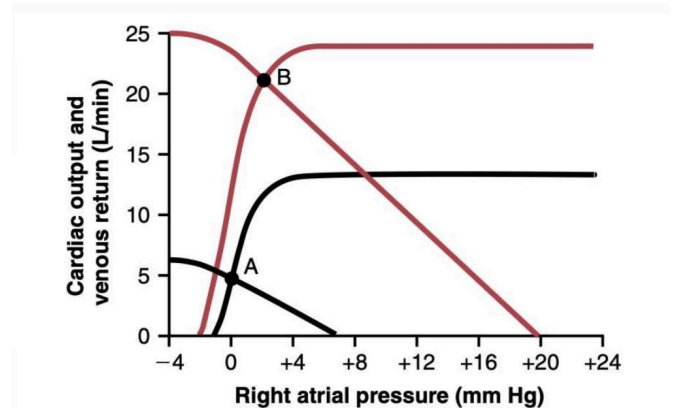


Figure 21-2. Graphic analysis of change in cardiac output, venous return, and right atrial pressure with the onset of strenuous exercise. Black curves, Normal circulation. Red curves, heavy exercise.

- CO and VR curves intersect at equilibrium points:

- Point A = rest
- Point B = heavy exercise

- Increased CO requires changes in VR -> cannot occur without adjustments

- **RAP = 0 at rest**

TWO KEY ADJUSTMENTS DURING HEAVY EXERCISE

1.) Increased mean systemic filling pressure (MSFP):

- Caused by sympathetic venoconstriction + abdominal and skeletal muscle compression
- MSFP can rise from 7 mmHg to 30 mmHg, enhancing VR

2.) Increased slope of VR curve

- Caused by decreased resistance in active muscle vessels
- Decreased resistance results in easier VR and steeper slope of VR curve

- **Increased MSFP + venous resistance = enhanced VR**

VR AND CO CURVES — NEW EQUILIBRIUM (EXERCISE VS REST)

1.) Equilibrium shift

- Point A = rest
- Point B = heavy exercise (increased CO, minimal change in RAP)

2.) RAP

WEEK 7-9: INTEGRATIVE PHYSIOLOGY

- In healthy individuals, RAP rises only ~1.5 mmHg during heavy exercise
- In strong hearts, RAP may even fall below normal due to strong sympathetic stimulation
- In weak hearts, even moderate exercise can cause marked rise in RAP